

Information theory, distance matrix, and molecular branching

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Information theory, distance matrix, and molecular branching

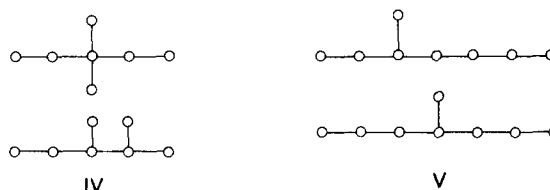
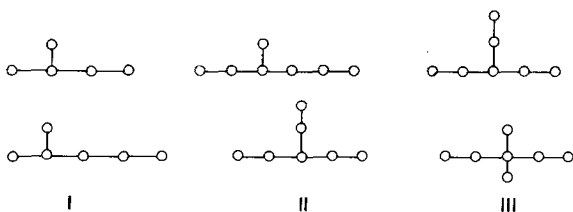
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Information theory was used in defining several measures of the topological properties of molecules, namely, information for adjacency, incidence, polynomial coefficients of the adjacency matrix, and for distances of molecular graphs. The latter was found to have a greater ability for discrimination between structural isomers than all known topological indices and to be a very appropriate measure of branching. All the quantities related to the distance matrix (two information measures, the Wiener number and the largest eigenvalue of the characteristic polynomial) were found to reflect in the same way the main features of molecular branching. On this basis, as well as on the basis of general expressions for information content and Wiener number derived for tree graphs, the essence of molecular branching was expressed in a number of rules. It was shown that these theoretical rules on branching are in agreement with the intuitive understanding of branching and are reflected in a large number of molecular properties.

I. INTRODUCTION

It is well known that molecular topology determines a large number of molecular properties.¹⁻⁵ Included here are not only those properties depending on the molecular size and shape such as boiling points, molecular volumes, refractive indices, etc., but also the thermodynamic and quantum mechanical characteristics of molecules.⁶⁻¹² In addition, it was found in the last few years that some biological activities of molecules, and even carcinogenicity, are closely related to molecular topology.¹³⁻¹⁵ Thus it is of pertinent interest for chemistry (as well as for other natural sciences) to have some quantitative measure reflecting the essential features of a given topological structure. Such measures are usually called topological indices. Several such indices have been suggested in the last 30 years.^{6,9,11,16-20} They have usually correlated more or less satisfactorily with the molecular properties but could not discriminate well between structural isomers, often providing the same index for different isomers. The reported correlations, however, reflect mainly the molecular properties' dependence on the number of atoms in a given series of compounds and not on the molecular branching in an individual molecule. It is also not completely clear (except perhaps intuitively) what the branching is and how it should be defined. For example, some authors^{19,20} prefer to consider as important for branching only the number of branched vertices. Others^{11,21} include here their valencies and, to some extent, the mode of distribution. However, no systematic study exists on the factors determining the branching: the length of the chain and branches, the total number of branches, the number of branches incident at the given vertex, the position of branches, etc. An explicit answer to this question means to discriminate in a unique way between the trees of the following type:



In this paper we examine the ability of information theory to provide quantitative measures of molecular topology having greater discriminatory power regarding the structural isomers than the known topological indices, and thus allowing a number of rules on the essence of molecular branching to be defined. The connection between these rules on branching, our intuitive understanding of branching, and some molecular properties will also be discussed.

Information theory²²⁻²⁵ is a useful tool in describing different structures. It provides a simple quantitative measure called information content of a given system. The information content of a system, I , having N elements is defined by the relation²³

$$I = N \log_2 N - \sum_{i=1}^n N_i \log_2 N_i, \quad (1)$$

where n is the number of different sets of elements, N_i is the number of elements in the i th set of elements, and summation is over all sets of elements. The logarithm is taken at basis 2 for measuring the information content in bits.

Another information measure is the mean information content of one element of the system, $\bar{I}$, defined by means of the total information content or by the Shannon relation²²:

$$\bar{I} = \frac{I}{N} = - \sum_{i=1}^n p_i \log_2 p_i, \quad (2)$$

where $p_i = N_i/N$. The application of information theory to different systems or structures is based on the possibility of constructing a finite probability scheme for every system:

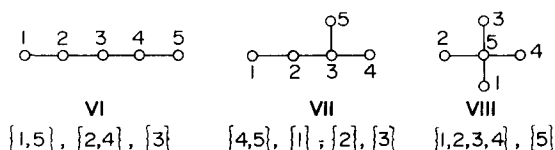
$$\begin{bmatrix} 1 & 2 & 3 & \cdots & n \\ N_1 & N_2 & N_3 & \cdots & N_n \\ p_1 & p_2 & p_3 & \cdots & p_n \end{bmatrix} \quad (3)$$

Considering all the N elements partitioned into n groups, we can define the probability p_i for a randomly selected element of this system to be found in the i th group. The criterion for partitioning the elements of a given system is not unique. The number of information measures is equal to the number of ways in which a set of N elements may be partitioned into different subsets, that is, the number of Young diagrams²⁶ for a given N . It is always possible to select for any system several information measures, each of them closely connected with certain properties of the system. The quantities so defined are statistical characteristics of the systems under study. They reflect the essence of the idea of information, defined by Ashby²⁴ as a measure of the variety in a given system, and by Glushkov²⁵ as a measure for the nonhomogeneity in distribution matter or energy into space or time. Thus the information content of a given system is zero in the case of full homogeneity, i.e., when all the elements form only one group [$N_1 = N$ in Eq. (1)]. It is maximal when each of the elements forms a separate group ($N_i = 1$ and $I_{\max} = N \log_2 N$).

II. BASIC DEFINITIONS AND CONCEPTS

Information theory was first applied to graphs in 1955 by Rashevsky,²⁷ who defined the so-called topological information content of the graph I_{top} . His definition is based on the partitioning of the vertices of a given graph into classes of equivalent vertices having the same valencies. Trucco²⁸ in 1956 made this definition more precise on the basis of an automorphism group of the graphs. In the latter case, two vertices are considered equivalent if they belong to the same orbit of the automorphism group, i.e., if they can interchange preserving the adjacency of the graph. In a series of papers, Mowshovitz²⁹ has studied in detail the information content of the graphs and digraphs and has formulated, on the basis of the chromatic properties of the graphs, the so-called chromatic information content of graphs. Bonchev *et al.*³⁰ introduced an information content of the molecules on the basis of point groups of symmetry and analyzed the relations between the information for topology and symmetry of molecules.

Topological information was applied by Rashevsky³¹ in studying the possibility of self-generation of the life on earth. Karreman³² has shown that the information balance of the chemical reactions can be positive, as well as negative, depending on the structure of reactants and products. The topological information measures, however, are not suitable for describing branching properties of graphs since they cannot reflect the essence of branching. This may be exemplified by considering trees with five vertices:



The five vertices are partitioned in different orbits in the above three graphs:

$$\text{VI } (2, 2, 1), \quad \text{VII } (2, 1, 1, 1), \quad \text{VIII } (4, 1).$$

Using Eq. (2), the following values for the information content in bits are obtained: $I_{\text{VI}} = 7.61$; $I_{\text{VII}} = 9.61$; $I_{\text{VIII}} = 3.61$. One can see that the above topological information (in Rashevsky's scheme) cannot reproduce the obvious fact that the branching increases from a chain, through a branched tree, to a star. On the other hand, the chromatic information measure has a rather limited number of possibilities for application to the chemical structures which are usually treated in connection with molecular branching.

One should note here that the term "topological information," used in the literature for the information on the orbits of the automorphism group of the graph (or shortly, orbits' information) may lead to some confusion. The group properties of a given graph are only a part of the properties of this topological structure. In the following, we shall define several other kinds of information content of a graph in order to find the most appropriate measure for molecular branching.

One should start looking for a possible measure among the various invariants of the graphs. These are, for instance, the adjacency matrix A and the incidence matrix L of the graph.³³ Let us consider these matrices for the graph (chain) VI:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The matrix elements a_{ij} and l_{ij} , composed solely of two values, form two sets of elements only: 25 (8, 17) and 20 (8, 12), respectively. One can calculate from here the probability of a randomly chosen matrix element in a_{ij} or l_{ij} to signify the adjacency or incidence, respectively ($p_A = \frac{8}{25}$ and $p_L = \frac{8}{20}$). In such a way, using Eqs. (1) and (2), one can define information for adjacency and information for incidence of a graph (I_{adj} and I_{inc}). These information quantities are constant for a given number of vertices N , since the number of matrix elements signifying the adjacency or incidence is $2(N-1)$, and hence

$$I_{\text{adj}} = N^2 \log_2 N^2 - 2(N-1) \log_2 2(N-1) - (N^2 - 2N + 2) \log_2 (N^2 - 2N + 2), \quad (4)$$

$$I_{\text{inc}} = N(N-1) \log_2 N(N-1) - 2(N-1) \log_2 2(N-1) - (N-1)(N-2) \log_2 (N-1)(N-2). \quad (5)$$

I_{adj} and I_{inc} are insensitive to any kind of branching and can distinguish only between trees and different cyclic structures. Nevertheless, they could be of use in the analysis of some problems connected with the adjacency and incidence of the graphs.

Another invariant of the graph is its characteristic polynomial.² Recently, in a number of papers, the

problem of molecular branching was related to the properties of the characteristic polynomial. Hosoya and co-workers^{9,18,34} have introduced a topological index Z which, in the case of acyclic molecules, is equal to the sum of the absolute values of the polynomial coefficients. Lovasz and Pelikan²⁰ have found that the maximal eigenvalue of a tree graph is a rather reliable measure of branching.

One can define an information content for polynomial coefficients $p(G, k)$ of a given graph by considering them as elements of a finite probability scheme. Then $p(G, k)/Z$ is a probability of a randomly chosen polynomial coefficient having a given value $p(G, k)$. Associating a certain probability to every coefficient in such a way, the information content of the graph I_{pc} can be calculated by Eqs. (1) and (2):

$$I_{pc} = Z \log_2 Z - \sum_{k=0}^{[N/2]} p(G, k) \log_2 p(G, k), \quad (6)$$

$$\bar{I}_{pc} = - \sum_{k=0}^{[N/2]} \frac{p(G, k)}{Z} \log_2 \frac{p(G, k)}{Z}. \quad (7)$$

[$N/2$ in Gauss' square brackets, see Eq. (6) above, is the greatest integer not exceeding the real number in them.] In this way one can, for instance, obtain for the information of polynomial coefficient of Graphs VI, VII, and VIII the following values:

$$\text{VI: } z=8, \quad p(G, 0)=1, \quad p(G, 1)=4, \quad p(G, 2)=3;$$

$$I_{pc} = 11.25 \text{ bits,}$$

$$\text{VII: } z=7, \quad p(G, 0)=1, \quad p(G, 1)=4, \quad p(G, 2)=2;$$

$$I_{pc} = 9.65 \text{ bits,}$$

$$\text{VIII: } z=5, \quad p(G, 0)=1, \quad p(G, 1)=4,$$

$$I_{pc} = 3.61 \text{ bits.}$$

This reflects well the change in the branching from the chain to the star.

It can be shown that the information for polynomial coefficients has the same behavior as Hosoya's topological index Z , being much more sensitive than the latter in distinguishing isomers of a given graph, since the same Z can be obtained from different values of $p(G, k)$. Nevertheless, this information measure is not a very powerful one since the number of the different decompositions of Z into $p(G, k)$ is limited by the small number of $k (=0, 1, \dots, [N/2])$. Inspection of the Hosoya tables of topological indices³⁴ Z shows that at $N=8$, from four pairs of graphs having the same index ($Z=29, 28, 25$, and 17), only the last one could not be distinguished by I_{pc} . At $N=9$, from five such pairs ($Z=48, 45, 40, 36$, and 31) only the underlined remained undistinguished.

Another invariant of the graph is its distance matrix D which contains the shortest paths (the distances d_{ij}) between every pair of connected vertices. Branching is connected with the distance matrix in an obvious way, since with increasing branching the distances in the graph become smaller. This can easily be seen from the distance matrices of the graphs VI, VII, and VIII:

$$D(\text{VI}) = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 1 & 0 & 1 & 2 & 3 \\ 2 & 1 & 0 & 1 & 2 \\ 3 & 2 & 1 & 0 & 1 \\ 4 & 3 & 2 & 1 & 0 \end{bmatrix}, \quad D(\text{VII}) = \begin{bmatrix} 0 & 1 & 2 & 3 & 3 \\ 1 & 0 & 1 & 2 & 2 \\ 2 & 1 & 0 & 1 & 1 \\ 3 & 2 & 1 & 0 & 2 \\ 3 & 2 & 1 & 2 & 0 \end{bmatrix},$$

$$D(\text{VIII}) = \begin{bmatrix} 0 & 2 & 2 & 2 & 1 \\ 2 & 0 & 2 & 2 & 1 \\ 2 & 2 & 0 & 2 & 1 \\ 2 & 2 & 2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}.$$

Wiener⁶ and Platt¹⁶ first made use of the connection between the distance matrix and branching defining a path number W which is equal to the sum of all the edges between all pairs of carbon atoms in hydrocarbons. The Wiener number is, in fact, the half-sum of the off-diagonal elements of a distance matrix. Hosoya¹⁸ has extended Wiener's definition to ring compounds, whereas Rouvray³⁵ has found a fairly good correlation between this topological index and the number of properties of alkanes.

One can introduce an information on distances in the graph I_D , considering all the matrix elements of distance matrix d_{ij} as elements of a finite probability scheme associated with the graph in question. Let the distance of a value i appear $2k_i$ times in the distance matrix, where $1 \leq i \leq N-1$, N being the number of vertices of the graph. Then N^2 matrix elements d_{ij} will be partitioned into $m+1$ groups, where m is the highest value of i , and $m+1$ group will contain N zeros which are the diagonal matrix elements. With each one of these $(m+1)$ groups can be associated a certain probability for a randomly chosen distance d_{ij} to be in the i th group:

$$p_i = 2k_i/N^2, \quad p_0 = N/N^2 = 1/N,$$

Distances:	0	1	2	3	$\dots$	m
Number of distances:	N	$2k_1$	$2k_2$	$2k_3$	$\dots$	$2k_m$
Probability:	$\frac{1}{N}$	$\frac{2k_1}{N^2}$	$\frac{2k_2}{N^2}$	$\frac{2k_3}{N^2}$	$\dots$	$\frac{2k_m}{N^2}$

The information on distances of a given graph will then, according to Eqs. (1) and (2), be

$$\bar{I}_D = - \frac{1}{N} \log_2 \frac{1}{N} - \sum_{i=1}^m \frac{2k_i}{N^2} \log_2 \frac{2k_i}{N^2}, \quad (8)$$

$$I_D = N^2 \log_2 N^2 - N \log_2 N - \sum_{i=1}^m 2k_i \log_2 2k_i. \quad (9)$$

Since D is a symmetric matrix, one can consider, for simplicity of discussion, only the upper triangular submatrix, which does indeed preserve all the properties of the information measure. In that case, we have the following expressions for the mean and total information on distances:

$$\bar{I}_D = -\sum \frac{2k_i}{N(N-1)} \log_2 \frac{2k_i}{N(N-1)}, \quad (8')$$

$$I_D = \frac{N(N-1)}{2} \log_2 \frac{N(N-1)}{2} - \sum k_i \log_2 k_i, \quad (9')$$

where $N(N-1)/2$ is the total number of upper off-diagonal elements in D . Using (8'), one can obtain the following information on distances in the three graphs with $N=5$:

VI	VII	VIII
$i \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ \frac{4}{10} & \frac{3}{10} & \frac{2}{10} & \frac{1}{10} \end{bmatrix}$	$i \begin{bmatrix} 1 & 2 & 3 \\ 4 & 4 & 2 \\ \frac{4}{10} & \frac{4}{10} & \frac{2}{10} \end{bmatrix}$	$i \begin{bmatrix} 1 & 2 \\ 4 & 6 \\ \frac{4}{10} & \frac{6}{10} \end{bmatrix}$
$\bar{I}_D = 1.85$	$\bar{I}_D = 1.52$	$\bar{I}_D = 0.97$

We see that $I_D = [N(N-1)/2] \bar{I}_D$ and $\bar{I}_D$ reproduce the branching properties of Graphs VI, VII, and VIII decreasing regularly with the increased branching. This is an interesting result since these information quantities are not closely connected with the Wiener number W ($W_{VI}=20$, $W_{VII}=18$, $W_{VIII}=16$). They are a statistical measure for the degree of nonhomogeneity in distribution distances in a given graph. I_D is a rather sensitive measure of branching having different values for all trees with $N=4, 5, 6, 7$, and 8 (see Table I).

The number of all the possible distributions of d_{ij} , i.e., the number of different I_D , increases rapidly with the increase of the number of vertices in the graph. This makes I_D an appropriate quantity for distinguishing structural isomers. However, there is another possible information measure which we can define on the

basis of distance matrix: *information on the realized distances* in a given graph, I_D^W . [In order to avoid possible confusion, we shall denote further the information defined in Eqs. (8')–(9') as I_D^E , since it is in the *information on the distribution of distances in the graph* according to their equality or nonequality.] I_D^W is closer to the Wiener number. It is information on the partitioning Wiener number (which is the total distance of the graph) into groups of distances of the same absolute values. Since the Wiener number is given by

$$W = \sum_i i k_i \quad (10)$$

and following Eqs. (1) and (2), we obtain

$$I_D^W = W \log_2 W - \sum_i k_i i \log_2 i \quad (11)$$

and

$$\bar{I}_D^W = -\sum_i k_i \frac{i}{W} \log_2 \frac{i}{W}. \quad (12)$$

For the three graphs with $N=5$, the following values are obtained:

VI:	$I_D^W = 62.93$,	$\bar{I}_D^W = 3.15$,
VII:	$I_D^W = 57.55$,	$\bar{I}_D^W = 3.20$,
VIII:	$I_D^W = 52.00$,	$\bar{I}_D^W = 3.25$.

One can see that I_D^W decreases with branching. It is a more sensitive quantity than the Wiener number since it can distinguish two graphs having the same W but different i and k_i . $\bar{I}_D^W$ increases regularly with branching at lower values of N , but at higher ones some irregularities occur and it cannot be used as a good measure of branching.

III. SOME ANALYTICAL EXPRESSIONS

As can be seen from Eqs. (10)–(12), it is necessary to have a mathematical expression defining k_i , the number indicating how many times a given distance i appears in the distance matrix of the graph. We have derived by induction the following formula for k_i of the upper triangle of the distance matrix, valid for any kind of branched trees except trees having branched branches:

$$k_i = k_i^{\text{chain}} + \sum_i k_i^{\text{branch}} + \sum_i k_i^{\text{branch/chain}} + \sum_i k_i^{\text{branch/branch}}, \quad (13)$$

$$k_i = N_0 - i + \sum_{j=2}^{N_0-1} \sum_{l=0}^{n_j} (N_{j1} - i) + \sum_{j=1}^{N_0} \sum_{j'=j}^{j+(i-1)} \left(n_{j'} - \sum_{d=0}^{i-1-\Delta_j-1} n_{j'} \right) + \sum_{j=2}^{N_0-1} \sum_{l=1}^{n_j-1} \sum_{l'=l+1}^{n_j} \delta_{ijl l'} + \sum_{j=2}^{N_0-2} \sum_{j'=j+1}^{N_0-1} \sum_{m=0}^{i-\Delta_j-2} \left(n_j - \sum_{d=0}^m n_j^d \right) \left(n_{j'} - \sum_{d=0}^{i-\Delta_j-2-m} n_{j'}^d \right), \quad (14)$$

where $\delta_{ijl l'}$ has the following values:

for $i \leq N_{j1}^{\min}$,	$\delta = i = 1$,
for $N_{j1}^{\min} < i < N_{j1} + N_{j1'}$,	$\delta = N_{j1}^{\min}$,
for $i = N_{j1} + N_{j1'}$,	$\delta = 1$,
for $i > N_{j1} + N_{j1'}$,	$\delta = 0$.

(15)

The following symbols are used in Eqs. (14) and (15): N_0 is the number of vertices in the chain; i is the distance between the two vertices; n is the total number of branches; n_j , $n_{j'}$ are the number of branches at the vertices j and j' , respectively; $\Delta_j = j' - j$; d is the length of the branch equal to the number of vertices in it N_{j1} where l enumerates the branches connected to the vertex j ; n_j^d is the number of branches with a given length d connected to the

vertex j ; n_j^0 is zero, by definition; N_{j1}^{min} is the length of the shortest branch incident to vertex j .

For hydrocarbons where more than two branches cannot be attached at one vertex, this equation reduces to simpler form: the fourth term in it will contain only the first sum and every sum in the third and fifth term will contain only one or two different numbers in the brackets.

One need not introduce Eq. (14) into (10) in order to define the Wiener number since the latter can be obtained easier in an inductive way by direct summation on the different parts of matrix D :

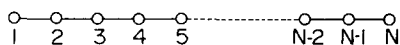
$$W = W^{\text{chain}} + \sum W^{\text{branch}} + \sum W^{\text{chain/branch}} + \sum W^{\text{branch/branch}}. \quad (16)$$

This relation can be given in more detailed form:

$$W = \frac{N_0(N_0^2 - 1)}{6} + \sum_{j=2}^{N_0-1} \sum_{i=1}^{n_j} \frac{N_{ji}(N_{ji}^2 - 1)}{6} + \sum_{j=2}^{N_0-1} \sum_{i=1}^{n_j} \frac{N_{ji}}{2} [(N_0 - j + 2)(N_0 - j + 1) + (j - 1)(j + 2) + N_0(N_{ji} - 1)] \\ + \sum_{i=1}^{n-1} \sum_{i'=i+1}^n \left(\frac{N_{ji} + N_{ji'} + 2}{2} + j_{i'} - j_i \right) N_i N_{i'}. \quad (17)$$

In the last term of (17) all the branches have a common enumeration $l = 1 - n$. Equations (10)–(15) and (17) allow a number of simple relations and inequalities to be derived for the information on distances in the graph and for Wiener number for those cases which are of interest in chemistry, as well as for a better understanding of the main features of molecular branching. Since the information on distances I_D and Wiener number reflect in the same way the principal features of branching, we shall give further diverse relations for Wiener number mainly because of their greater simplicity.

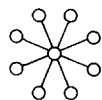
Chain



$$W = N(N^2 - 1)/6, \quad (18)$$

$$I_D^E = \frac{N(N-1)}{2} \log_2 \frac{N(N-1)}{2} - \sum_{k=1}^{N-1} k \log_2 k. \quad (19)$$

Star: $N_0 = 3$, $j = 2$, $N_i = 1$



$$W = (N-1)^2, \quad (20)$$

$$I_D^E = \frac{N(N-1)}{2} \log_2 \frac{N(N-1)}{2} - (N-1) \log_2 (N-1) \\ - \frac{(N-1)(N-2)}{2} \log_2 \frac{(N-1)(N-2)}{2}. \quad (21)$$

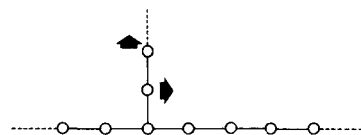
Tree with one branch of a minimal length: $n = 1$, $N_i = 1$, $j \neq \text{const}$



$$W = \frac{N_0(N_0^2 - 1)}{6} + \frac{1}{2} [(N_0 - j + 2)(N_0 - j + 1) + (j - 1)(j + 2)]. \quad (22)$$

$$I_D^E = \frac{N(N-1)}{2} \log_2 \frac{N(N-1)}{2} - (N-1) \log_2 (N-1) \\ - \sum_{k=1}^{N-1} k \log_2 k \quad (k \neq j-1, N-j). \quad (23)$$

Tree with one branch of an arbitrary length: $n = 1$, $N_i > 1$, $j \neq \text{const}$



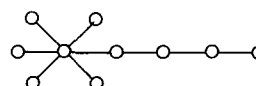
$$W = \frac{N_0(N_0^2 - 1)}{6} + \frac{N_1(N_1^2 - 1)}{6} + \frac{N_1}{2} [(N_0 - j + 2)(N_0 - j + 1) + (j - 1)(j + 2) + N_0(N_1 - 1)] \quad (\text{for } N_1 = 2), \quad (24a)$$

$$I_D^E = \frac{N(N-1)}{2} \log_2 \frac{N(N-1)}{2} - 2(N-1) \log_2 (N-1) - \sum_{k=1}^{N-1} k \log_2 k, \quad (24b)$$

where

$$k \neq j-2, j-1, N-j, N-j+1, \text{ for } j = \frac{N-N_i+1}{2}, \\ k \neq j-2, j, N-j+1, \text{ otherwise.}$$

Tree with several branches of a minimal length all attached to only one vertex: $n = n_j$, $N_{ji} = 1$

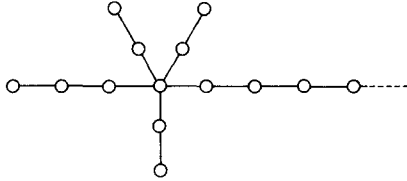


$$W = \frac{N_0(N_0^2 - 1)}{6} + \frac{n}{2} [(N_0 - j + 2)(N_0 - j + 1) + (j - 1)(j + 2) + n(n-1)] \quad (\text{for } n = 2),$$

$$I_D^E = \frac{N(N-1)}{2} \log_2 \frac{N(N-1)}{2} - (N+1) \log_2 (N+1) - (N-1) \log_2 (N-1) - \sum_{k=1}^{N-1} k \log_2 k, \quad (25)$$

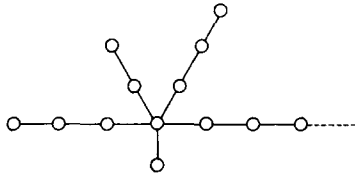
$$(k \neq j-1, n-j, N-j+1).$$

Tree with several branches of an arbitrary but equal length all attached to only one vertex: $n = n_j$, $N_{ji} > 1$, $N_{ji} = \text{const}$



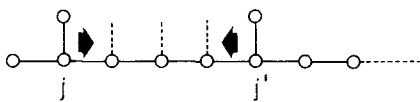
$$W = \frac{N_0(N_0^2-1)}{6} + n \frac{N_i(N_i^2-1)}{6} + n \frac{N_i}{2} [(N_0-j+2)(N_0-j+1) + (j-1)(j+2) + N_0(N_i-1)] + \frac{n(n-1)}{2} N_i^2(N_i+1). \quad (26)$$

Tree with several branches of an arbitrary but different length all attached to only one vertex



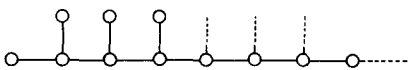
$$W = \frac{N_0(N_0^2-1)}{6} + \sum \frac{N_i(N_i^2-1)}{6} + \frac{1}{2} \sum_{i=1}^n N_i [(N_0-j+2) \times (N_0-j+1) + (j-1)(j+2)] + \frac{N_0}{2} \sum_{i=1}^n N_i(N_i-1) + \frac{1}{2} \sum_{i=1}^{n-1} \sum_{i'=i+1}^n N_i N_{i'} (N_i + N_{i'} + 2). \quad (27)$$

Tree with two branches of a minimal length attached to different vertices: $n = 2$, $n \neq n_j$, $N_j = 1$, $j, j \neq \text{const}$



$$W = \frac{N_0(N_0^2-1)}{6} + \frac{1}{2} [(N_0-j_k+2)(N_0-j_k+1) + (j_k-1)(j_k+2)] + 2 + (j_{k'}-j_k) \quad (k' > k). \quad (28)$$

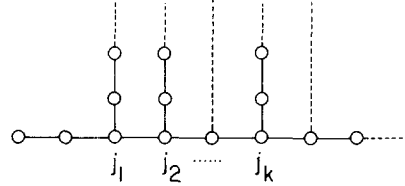
Tree with several branches with a minimal length connected with different vertices: $n > 1$, $n \neq n_j$, $N_{ji} = 1$



$$W = \frac{N_0(N_0^2-1)}{6} + \frac{1}{2} \sum_{k=1}^n [(N_0-j_k+2) + (j_k-1)(j_k+2)]$$

$$+ n(n-1) + \sum_{k=1}^{n-1} \sum_{k'=k+1}^n (j_{k'}-j_k). \quad (29)$$

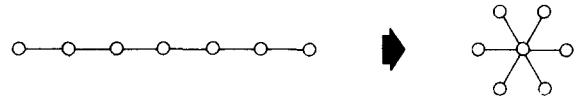
Tree with several branches with an equal but not minimal length, connected to different vertices: $n > 1$, $N_i = \text{const} > 1$, $n \neq n_j$



$$W = \frac{N_0(N_0^2-1)}{6} + n \frac{N_i(N_i^2-1)}{6} + \sum_{i=k+1}^n \frac{N_i}{2} [(N_0-j_k+2) \times (N_0-j_k+1) + (j_k-1)(j_k+2) + N_0(N_i-1)] + \sum_{i=k+1}^{n-1} \sum_{k'=k+1}^n N_i^2 [(N_i+1) + (j_{k'}-j_k)]. \quad (30)$$

Some inequalities concerning branching ($N = \text{const}$)

Chain-star



$$\Delta W = \frac{1}{6} (N-1)(N-2)(N-3) \geq 0, \quad (31)$$

$$\Delta I_D^E = \sum_{k=1}^{N-2} k \log_2 \sum_{k=1}^{N-2} k - \sum_{k=1}^{N-2} k \log_2 k \geq 0. \quad (32)$$

The equality occurs only at $N \leq 3$, where there is no difference between a chain and a star. Thus

$$W^{\text{chain}} > W^{\text{star}}, \quad I_D^{\text{chain}} > I_D^{\text{star}}. \quad (33)$$

Chain-branched tree



It is sufficient to take a tree having one branch with a minimal length [Eq. (22)]:

$$\Delta W = (N_0-j)(j-1) > 0 \quad (j=2-N/2 \text{ or } N/2+1), \quad (34)$$

$$\Delta I_D^E = (N-1) \log_2 (N-1) - j \log_2 j - (N-j-1) \log_2 (N-j-1) > 0, \quad (35)$$

since $(x+y) \log_2 (x+y) > \log_2 x + y \log_2 y$.

These inequalities appear stronger with the increase of j , i. e., when the branch moves from the end to the middle of the chain. Thus,

$$W^{\text{chain}} > W^{\text{branched tree}}, \quad I_D^{\text{chain}} > I_D^{\text{branched tree}}. \quad (36)$$

Branched tree-star

From the inequalities (42)–(48) it immediately follows that all kinds of branched trees can be reduced to star

with a decrease of the Wiener number W . Then

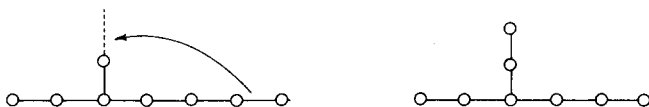
$$W^{\text{tree}} > W^{\text{star}} \quad (37)$$

One can now unite inequalities (33), (36), and (37) to a common one:

$$W^{\text{chain}} > W^{\text{branched tree}} > W^{\text{star}}, \quad (38)$$

$$I_D^{\text{chain}} > I_D^{\text{branched tree}} > I_D^{\text{star}} \quad (39)$$

One-branched trees with a different branch length



I. $N = \text{const}$, N_0, N_1 II. $N'_0 = N_0 - 1$, $N'_1 = N_1 + 1$

$$\Delta W = (j-1)(N_0 - N_1 - j - 1) > 0. \quad (40)$$

The inequalities are in force at $j > 1$ (a trivial requirement for increasing the length of the branch), and at $N_0 > N_1 + j + 1$. In the last case

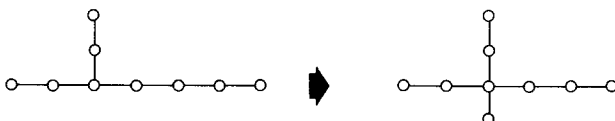
$$j_{\max} = \begin{cases} \frac{N_0}{2} & \text{if } N_0 = 2k \\ \frac{N_0}{2} + \frac{1}{2} & \text{if } N_0 = 2k + 1 \end{cases}$$

$$\text{hence } \begin{cases} N_0 > 2N_1 + 2, \text{ or } N_1 < N_0/2 - 1 \\ N_0 > 2N_1 + 3, \text{ or } N_1 < N_0/2 - 1.5 \end{cases}$$

If equality instead of inequality occurs, the new graph will be identical with the initial one. If the opposite inequality occurs, the branch will be greater than the one side chain and Δw will decrease. Then if we restrict the length of branches to be less than any side chain, the following inequality is in force:

$$W_{N_0, N_j}^{\text{tree}} > W_{N_0-1, N_{j+1}}^{\text{tree}} \quad (41)$$

Trees with a different number of branches attached to one vertex



$N = \text{const}$

$j = \text{const}$ $n_1 = 1$, N_1, N_0 $n_2 = 2$, $N_2 = 1$, $N'_0 = N_0 - 1$,

$$\Delta W = (N_0 - j - 1)(N_1 + j - 1) > 0. \quad (42)$$

Since $j, N_1 > 0$, $N_1 + j - 1 > 0$, and since $2 \leq j \leq (N_0 + 1)/2$, $N_0 - j + 1 > 0$ at $N_0 > 3$, i.e., at every tree, then

$$W_{n_j, N_0}^{\text{tree}} > W_{n_{j+1}, N_0-1}^{\text{tree}} \quad (43)$$

Trees having a branch formed by decreasing the length of a long branch attached to the same vertex



$N_0 = \text{const}$, $j = \text{const}$, $n = 1$, $N_1 > 1$,

$n' = 2$, $N_2 = 1$, $N'_1 = N_1 - 1$

$$\Delta W = (N_1 - 1)(N_0 - 1) > 0. \quad (44)$$

The inequality is in force for any $N_1 > 1$. At a given N_0 , the greater N_1 , the stronger the inequality. Then,

$$W_{n_j, N_j}^{\text{tree}} > W_{n_{j+1}, N_{j-1}}^{\text{tree}} \quad (44')$$

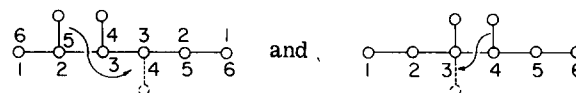
Trees having an equal number of branches with a same length attached to one or different vertices, respectively



$N_0, n, N_i = \text{const}$, $n_j = 1$, $n'_j = 1$, $n_j = 2$, $n_{j'} = 0$

$$\Delta W = (j' - j)(j' + j - N_0). \quad (45)$$

Let us mention that in all our equations $j' > j$. To avoid some possible confusion in applying inequality (45), let us consider the trees



We see that at a normal left-to-right enumeration of the chain vertices in the first graph we have $j' < j$. To avoid such situations, one should use a reverse (right-to-left) enumeration. In that case, $j' = 5$, $j = 4$ will provide correct values of ΔW .

Then because of $j' > j$, the first term in (45) will always be positive. Hence, ΔW will be positive, zero, or negative when $j + j'$ is greater than, equal to, or less than N_0 .

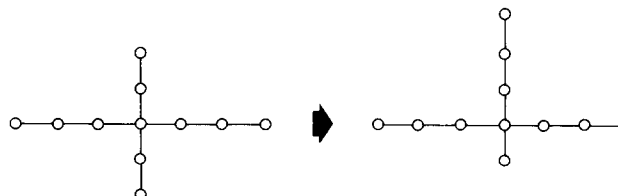
If we introduce an additional requirement, that the two branches be in equivalent positions ($j' = N_0 - j + 1$), Eq. (45) will be simplified to

$$\Delta W = j' - j > 0. \quad (46)$$

Then

$$W_{n_j, n_{j'}}^{\text{tree}} > W_{n_{j+1}, n_{j'-1}}^{\text{tree}} \quad (46')$$

Trees having two branches with an arbitrary length attached to the same vertex



$N_0, n, j = \text{const}$, N_1, N_2 , $N_1 \geq N_2$, $N_1 + 1$, $N_2 - 1$,

$$\Delta W = \frac{1}{2}(N_2 - N_1)[(N_1 + N_2)(N_1 + 1) + N_0 + 2N_1] + 1 - N_0 < 0. \quad (47)$$

Then

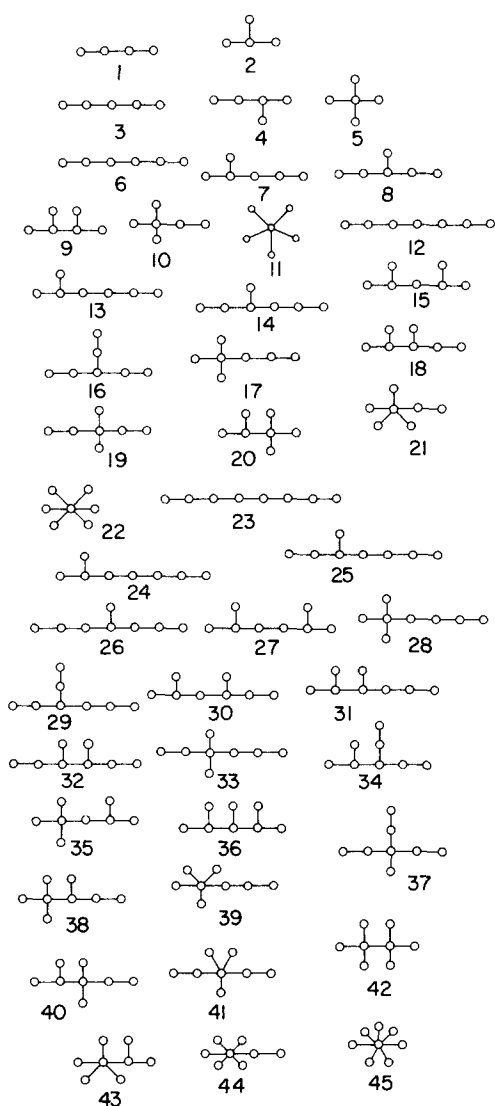


FIG. 1. Diagrams of all tree graphs with four, five, six, seven, and eight vertices.

$$W_{N_k, N'_k}^{\text{tree}} < W_{N_k+1, N'_k-1}^{\text{tree}} \quad (47')$$

Trees having a different number of branches of a minimal length



$$\begin{aligned} N_0, n, j_1, j_2, \dots, j_k, N_j=1 \\ N_0+1, n-1, j_1, j_2, \dots, j_{k-1} \\ \Delta W = (j_k - N_0)(n + j_k - 2) < 0, \end{aligned} \quad (48)$$

where j_k is the vertex number of the disappearing branch. Then

$$W_{N_0, n(n_i=1)}^{\text{tree}} > W_{N_0-1, n+1(n_i=1)}^{\text{tree}} \quad (49)$$

IV. NUMERICAL RESULTS. COMPARISON BETWEEN THE VARIOUS BRANCHING INDICES

The values of both information measures, introduced on the basis of distance matrices, were calculated for

TABLE I. Total branching indices of tree graphs having four–eight vertices.

Graph ^a	W	I_D^E	I_D^W	χ_1	Z	I_{pc}	χ_R
N = 4							
1.	10	8.7549	24.4644	1.618	5	6.8547	1.9142
2.	9	6.	22.5293	1.732	4	3.2451	1.7321
N = 5							
3.	20	18.4644	62.9288	1.732	8	11.2451	2.4142
4.	18	15.2193	57.5488	1.848	7	9.6515	2.2701
5.	16	9.7095	52.0000	2.000	5	3.6096	2.
N = 6							
6.	35	32.2389	129.6506	1.802	13	20.9863	2.9142
7.	32	28.6293	119.7353	1.902	11	14.8345	2.7702
8.	31	27.3842	116.5605	1.932	12	19.8003	2.8082
9.	29	23.4840	109.8618	2.000	10	13.6097	2.6425
10.	28	22.5874	106.3412	2.074	9	12.1648	2.5607
11.	25	13.7745	96.0964	2.236	6	3.9002	2.2361
N = 7							
12.	56	50.3644	233.4633	1.848	21	35.5096	3.4142
13.	52	46.4642	218.1841	1.932	18	29.0195	3.2700
14.	50	44.8546	210.5087	1.970	19	31.9166	3.3081
15.	48	41.0774	203.0586	2.000	15	19.0936	3.1251
16.	48	40.9544	203.5488	2.000	20	34.3995	3.3461
17.	46	39.9740	195.0642	2.101	14	18.1417	3.0607
18.	46	39.5676	195.5544	2.053	17	27.9771	3.1807
19.	44	37.2191	187.6856	2.136	16	26.8387	3.1213
20.	42	32.6898	179.9479	2.175	13	17.0862	2.9432
21.	40	30.6752	171.8575	2.288	11	14.5439	2.8121
22.	36	18.1255	156.1173	2.449	7	4.1417	2.4495
N = 8							
23.	84	73.0801	403.3317	1.879	34	61.4995	3.9142
24.	79	68.9384	362.3758	1.950	29	48.2755	3.7701
25.	76	67.0382	349.9745	1.989	31	56.6256	3.8081
26.	75	66.1835	345.9030	2.000	30	50.2522	3.8081
27.	74	63.3448	341.2866	2.000	25	40.3392	3.6259
28.	71	62.0606	328.0287	2.112	23	36.6160	3.5607
29.	72	62.0418	333.7311	2.028	32	58.5162	3.8510
30.	71	61.8350	328.8834	2.042	26	42.8448	3.6639
31.	70	61.3029	324.5463	2.074	27	45.1150	3.6807
32.	68	58.9544	316.2987	2.095	29	53.4727	3.7188
33.	67	58.7736	311.5341	2.157	25	41.8158	3.6213
34.	67	55.3448	312.3888	2.101	28	47.1972	3.7188
35.	66	54.6157	307.1555	2.149	19	23.0054	3.4165
36.	65	54.4251	303.4147	2.136	24	39.3681	3.5534
37.	64	53.1409	299.2059	2.189	28	52.2834	3.6819
38.	63	52.9802	294.5294	2.206	22	35.6474	3.4814
39.	62	52.3253	289.3857	2.303	17	21.3061	3.3121
40.	62	51.2058	290.3661	2.222	23	38.3367	3.5040
41.	59	47.9349	277.0367	2.334	20	33.5029	3.3882
42.	58	43.4056	272.9688	2.303	17	21.3061	3.2500
43.	57	42.8487	268.4355	2.358	16	20.3485	3.4612
44.	54	39.3448	254.9894	2.488	13	16.8446	3.0367
45.	49	22.7157	233.1208	2.646	8	4.3485	2.6458

^aNumbers correspond to the tree graphs given in Fig. 1.

the tree graphs having four, five, six, seven, and eight vertices (Fig. 1). These results are given in Table I in comparison with several other branching indices which have relatively good power of discrimination between isomers. Included here are the Wiener number⁶ W , the greatest eigenvalue of the characteristic polynomial,^{20,36} χ_1 , the sum of the polynomial coefficients (or Hosoya index)¹⁸ Z , the information on polynomial coefficients, I_{pc} , and Randić connectivity index¹¹ χ_R .

The inspection of Table I indicates the great sensitivity of the two information quantities (I_D^E , I_D^W) to all structural details of the tree graphs. There are no two graphs among the 45 graphs examined which have the same information on the graph distances. All the other listed indices are not so specific, and they often have

the same value for different graphs. Thus, for tree graphs having seven vertices, the Wiener number is the same for two pairs of trees (Nos. 15, 16 and 17, 18) while for x_1 , we have one such pair (Nos. 15 and 16). For trees with $N=8$, the Wiener number gives three identical pairs (Nos. 28, 30; 33, 34; and 39, 40); x_1 two pairs (Nos. 26, 27 and 39, 42); Randić's index, also two identical pairs (Nos. 25, 26 and 32, 34); and the Hosoya index gives five such pairs (Nos. 24, 32; 34, 37; 27, 33; 28, 40; and 39, 42). When considering larger trees ($N>8$), the number of these inconsistencies increased rapidly. Thus the information measures introduced on the basis of distance matrix appear to be very appropriate indices for discrimination of graphs. The number of different I_D^E for the graphs having the same number of vertices is limited by the number of all the possible distributions $N(N-1)/2$ graph edges into K different groups ($K=1, 0, \dots, [N-1]$), that is, the number of the possible Young diagrams of the tree graph. Since this number increases rapidly with increasing values of N , one may expect the information on graph distances to have a good ability of differentiation between structural isomers even for very large systems. The other information measure I_D^W is also valuable since here the number of possible different distributions appears as an additional discrimination possibility in the case of equal total distance W , though W usually has different values.

It seems very important to find out if a certain common basis exists between the different topological indices for a better understanding of the essence of branching. Figure 2 shows a very good correlation between both information measures, I_D^E and I_D^W , and the Wiener number W . The largest eigenvalue of the characteristic polynomial x_1 also correlates fairly well with these three measures, reproducing in the same manner

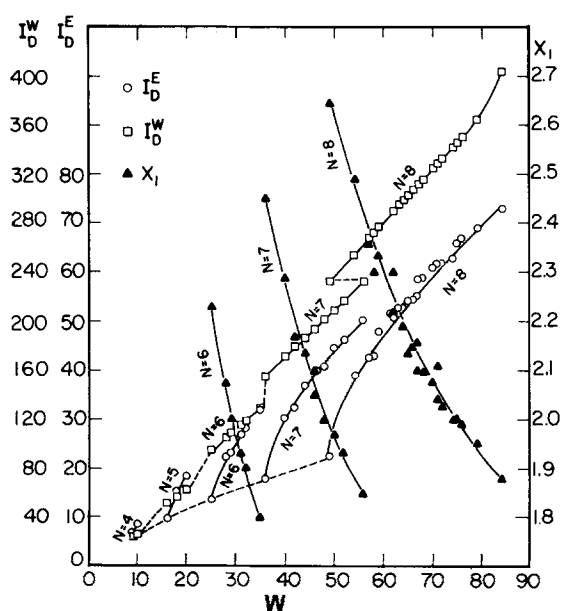


FIG. 2. Correlation between the information indices I_D^E and I_D^W , the Wiener number W , and the largest eigenvalue of the characteristic polynomial x_1 .

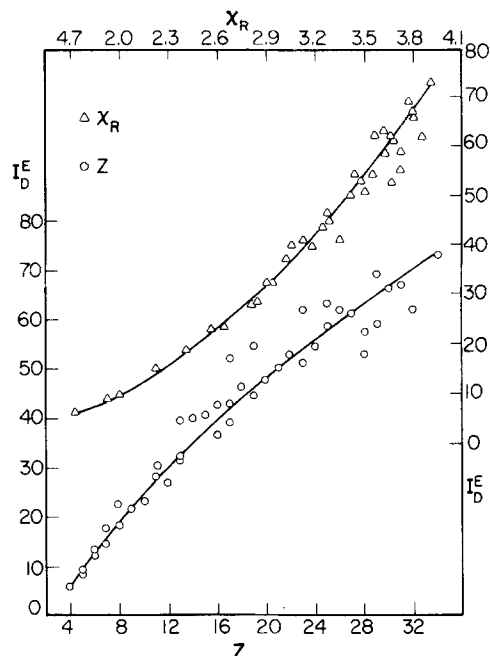
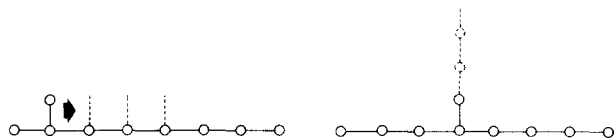


FIG. 3. Correlation of the information index I_D^E vs the branching indices Z and x_R .

the mean branching features of the tree graphs. This correlation between graph spectral characteristics and branching may be understood in light of the quite recent finding of Cvetković and Gutman³⁷ that the largest eigenvalue x_1 is approximately connected with the total number of walks in the graph. Thus the four closely correlated quantities have a common basis—the distance matrix of the graph—and they reproduce the essential features of branching at a given number of vertices N in a similar way. They can also reflect the change in the molecular skeleton when, at a given kind of branching, the number of vertices increases. In Fig. 2, this is illustrated by the dotted curve connecting graphs of the star type only. Other such relationships can easily be found in Fig. 2, for instance for chain graphs, for graphs having only one terminal branch, etc. All these properties of the four quantities, related through a distance matrix, could be used as a basis for defining the essence of branching in several rules. Such a set of rules will be given in the next section.

In Fig. 3, the correlation between the information on graph distances I_D^E and branching indices of Hosoya (Z) and Randić (x_R) is shown. In this way, the basis for defining branching rules is enlarged. Some of these rules will be covered by topological indices.

The deviations from the mean curve, however, reflect two very important differences between the two information measures, Wiener number and the largest graph spectrum eigenvalue on one side and Hosoya and Randić indices on the other side. These two groups of quantities give a different answer to the question of how the branching changes (at a given number of vertices) with the change in (i) the branch position, and (ii) the branch length:



We shall demonstrate later that the first group of branching indices gives an answer to the above question which is close to our intuitive understanding of branching.

The problem of understanding what the branching is requires a procedure (or some system of rules) for distinguishing between differently branched tree graphs having the same number of vertices, as well as having different number of vertices. The latter case is also worthwhile to consider in detail. The principles of branching must be common and independent on the number of vertices of the graph, N . Although Fig. 2 shows certain possibilities for finding some branching rules for $N \neq \text{const}$, one can use another approach here. The comparison of branching of two graphs having different number of vertices is possible on the basis of some *relative (or mean) branching index*. Thus the mean distance in the graph, $\bar{W}$, could be introduced dividing the Wiener number W by the number of distances [which is $N(N-1)/2$] instead by N :

$$\bar{W} = \frac{2W}{N(N-1)}. \quad (50)$$

In a similar way, we can introduce four other relative branching indices:

$$\begin{aligned} \bar{I}_D^E &= I_D^E/m, \\ \bar{I}_D^W &= I_D^W/W, \\ \bar{Z} &= Z/K, \\ \bar{\chi}_R &= \chi_R/(N-1), \end{aligned} \quad (51)$$

where m and K are the different groups of equal distances and polynomial coefficients, respectively. $N-1$ is the number of bonds in the molecule. For the largest graph spectrum eigenvalue (χ_1) such a procedure does not seem reasonable.

The values of these five relative branching indices are given in Table II for the tree graphs having $N=4, 5, 6, 7, 8$. The mean distance in the graph, $\bar{W}$, has some interesting properties. For instance, very simple formulas exist for chain and star graphs:

$$\begin{aligned} \bar{W}_{\text{chain}} &= (N+1)/3 \\ \bar{W}_{\text{star}} &= 2(N-1)/N. \end{aligned} \quad (52)$$

This quantity, however, cannot be used for quantitative comparison of branching of the graphs having different number of vertices. When N increases, $\bar{W}$ also increases without displaying if the new vertices are added to the graph form chain, tree, or star graph. Randić's mean measure $\bar{\chi}_R$ decreases with N in the same way as $\bar{W}$ increases. The second mean information measure, $\bar{I}_D^W$, is also not appropriate since at $N=8$, the higher values of N , it changes the correct branching order. The same happens with the mean polynomial coefficient $\bar{Z}$ starting from $N=5$. The only mean quantity which is

TABLE II. Mean branching indices of tree graphs having four, five, six, seven, and eight vertices.

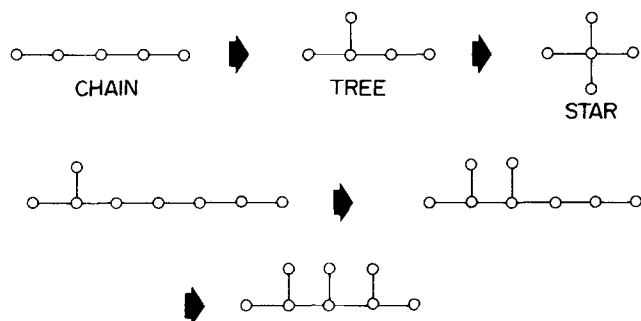
Graph ^a	$\bar{W}$	$\bar{I}_D^E$	$\bar{I}_D^W$	$\bar{Z}$	$\bar{I}_{pc}$	$\bar{\chi}_R$
$N=4$						
1.	1.6667	1.4592	2.4464	1.6667	2.2849	0.6381
2.	1.5	1.	2.5033	2.	1.6226	0.5774
$N=5$						
3.	2.	1.8464	3.1464	2.6667	3.7484	0.6034
4.	1.8	1.5219	3.1972	2.3333	3.2172	0.5675
5.	1.6	0.9710	3.2500	2.5	1.8048	0.5000
$N=6$						
6.	2.3333	2.1493	3.7043	3.25	5.2466	0.5828
7.	2.1333	1.9086	3.7417	3.6667	4.9448	0.5540
8.	2.0667	1.8256	3.7600	3.	4.9501	0.5616
9.	1.9333	1.5656	3.7883	3.3333	4.5366	0.5285
10.	1.8667	1.5058	3.7979	3.	4.0549	0.5121
11.	1.6667	0.9183	3.8439	3.	1.9501	0.4472
$N=7$						
12.	2.6667	2.3983	4.1690	5.5	8.8774	0.5690
13.	2.4762	2.2126	4.1958	4.5	7.2549	0.5450
14.	2.3810	2.1359	4.2102	4.75	7.9792	0.5514
15.	2.2857	1.9561	4.2304	5.	8.5999	0.5209
16.	2.2857	1.9502	4.2406	5.	6.0472	0.5577
17.	2.1905	1.9035	4.2405	4.6667	6.9943	0.5101
18.	2.1905	1.8842	4.2512	4.25	6.3645	0.5301
19.	2.0952	1.7723	4.2656	4.	6.7098	0.5202
20.	2.	1.5567	4.2845	4.3333	5.6954	0.4905
21.	1.9048	1.4607	4.2964	3.6667	4.8480	0.4687
22.	1.7143	0.8631	4.3366	3.5	2.0709	0.4083
$N=8$						
23.	3.	2.6100	4.8016	6.8	12.2999	0.5592
24.	2.8215	2.4621	4.5870	7.25	12.0689	0.5386
25.	2.7143	2.3942	4.6049	6.2	11.3251	0.5440
26.	2.6786	2.3637	4.6120	7.5	12.5631	0.5440
27.	2.6429	2.2623	4.6120	6.25	10.0848	0.5180
28.	2.5357	2.2165	4.6201	5.75	9.1540	0.5087
29.	2.5714	2.2158	4.6352	6.4	11.7032	0.5501
30.	2.5357	2.2084	4.6322	6.5	10.7112	0.5234
31.	2.5	2.1894	4.6364	6.75	11.2788	0.5258
32.	2.4286	2.1055	4.6515	5.8	10.6945	0.5313
33.	2.3929	2.0991	4.6498	6.25	10.4540	0.5173
34.	2.3929	1.9766	4.6625	7.	11.7993	0.5313
35.	2.3571	1.9506	4.6539	6.3333	7.6685	0.4881
36.	2.3214	1.9438	4.6679	6.	9.8420	0.5076
37.	2.2857	1.8979	4.6751	5.6	10.4567	0.5260
38.	2.25	1.8922	4.6751	5.5	8.9119	0.4973
39.	2.2143	1.8688	4.6675	5.6667	7.1020	0.4732
40.	2.2143	1.8288	4.6833	5.75	9.5842	0.5008
41.	2.1071	1.7120	4.6955	5.	8.3757	0.4840
42.	2.0714	1.5502	4.7064	5.6667	7.1020	0.4643
43.	2.0357	1.5303	4.7094	5.3333	6.7828	0.4945
44.	1.9286	1.4052	4.7220	4.3333	5.6149	0.4338
45.	1.75	0.8113	4.7576	4.	2.1743	0.3780

^aNumbers correspond to the tree graphs given in Fig. 1.

to any great extent convenient for estimations of the relative branching of graphs with different N is the mean information measure $\bar{I}_D^E$. It reflects nicely the main features of branching, including the increase of branching in stars and the decrease of branching in chains when N increases. Only in the case when the new vertex added turns chain into a tree with one branch could $\bar{I}_D^E$ not reflect the increase of the relative branching. It is our belief, however, that there is no function capable of reflecting *all* the features of branching in graphs having different numbers of vertices. Hence the mean information on the graph distances $\bar{I}_D^E$ could be of some use in studying branching in such graphs.

V. RULES ON BRANCHING

The discussions on the essence of branching are usually based on intuitive ideas only.³⁸ Intuition is a good guide in some simple cases.

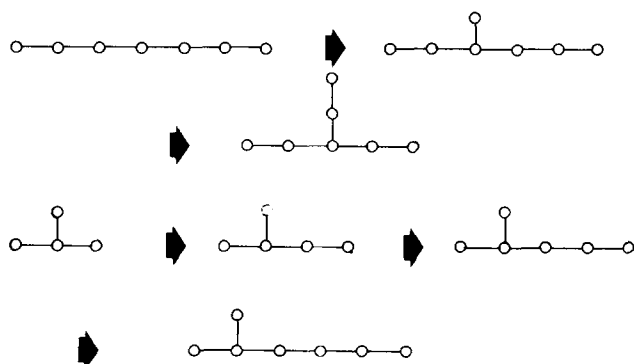


One can easily judge that the tree is a less branched graph than the star and that it is more branched than the chain. In the second case, it is also clear that the tree having the most branches is the most branched one. A large number of complicated graphs exist, however, where our intuition could hardly be of any use.

It is our opinion that a good approach to the problem: What is chain branching? consists in formulating certain general rules of branching. Such rules should be based on an appropriate mathematical formalism. They should also take into account all the factors influencing the branching and, simultaneously, they should remain as close as possible to the intuitive understanding of branching.

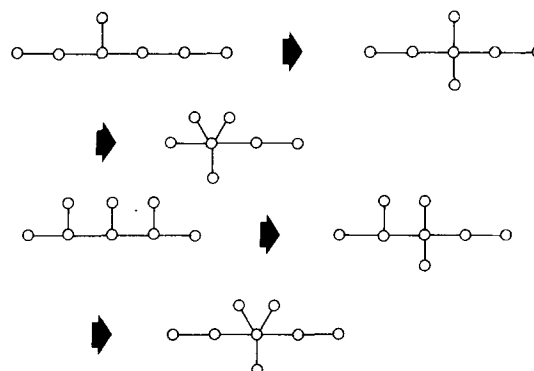
It is important to notice that our intuition provides many more details of branching than is usually supposed. The factors important in branching are (a) the total number of branches, (b) the length of the chain and branches, (c) the position of the branches, and (d) the number of branches attached to a given vertex. The two examples shown above demonstrate the influence of the most obvious factor, the total number of branches.

Let us now compare the following trees:



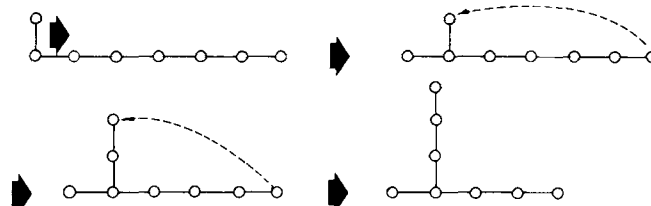
It is in accordance with our intuition to judge that in general the branching should increase when the length of branches increases. However, the branching should decrease relatively when the chain length increases.

Examining two other cases,

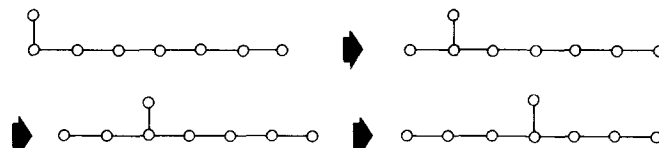


one could conclude that the branching increases with the increased number of branches connected to a given vertex. The second case, in fact, is not so obvious, but it could be associated with the tendency of forming a star-like graph which is more branched than the other ones.

The influence of the last factor, the position of branches, could be explained by the combined use of two other factors—the number of branches and their length:



Or otherwise:



In such a way, one can suppose that branching increases when the branch moves from the end to the middle of the chain.

All the examples considered above display the ability of our intuition to indicate the general tendency of branching to increase with the total number of branches, the length of branches, their more central position, and their number to a given vertex, and to decrease with the chain length increase. One could try to generalize this tendency in some simple rules on the basis of the quantities connected with the distance matrix (both information measures on graph distances, Wiener number, and the largest eigenvalue of the characteristic polynomial). In doing this we shall denote branching by the letter B . We shall also assume the branch length to be less than or equal to the half-length of the chain.

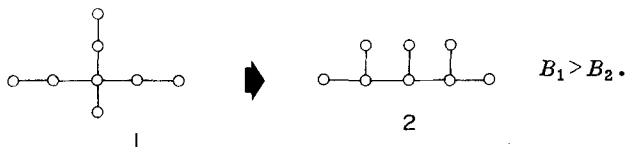
First of all we shall examine graphs having a constant number of vertices.

A. $N = \text{const}$

1. The influence of the total number of branches, n

In general, B tends to increase with n . The length and the position of the branches, N_{ji} and j , may cause

a graph with a smaller number of branches of various lengths attached to the same vertex to be more branched than a graph with a greater number of branches attached to different vertices as, for example,

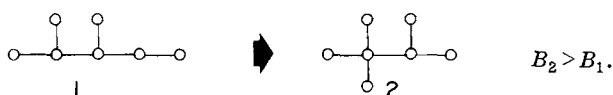


Keeping constant the position of branching, the following rules are formulated:

Rule 1.1. In a tree having a constant number of vertices, the branching increases when the number of branches attached to a given vertex increases owing to the chain decrease:

$$B_{n_j+1, N_0-1}^{\text{tree}} > B_{n_j, N_0}^{\text{tree}} \quad (53)$$

Here N_0 and $N_0 - 1$ are the number of the chain vertices, and n_j and $n_j + 1$ are the number of branches attached to the vertex j in the graphs with a longer and a shorter chain, respectively. It comprises trees having branches attached not only to one vertex but also to different vertices, as, for instance,



The two extreme cases of inequality (53) may be formulated as a special rule:

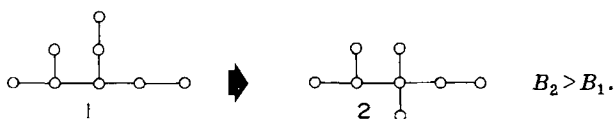
Rule 1.2. A tree is always more branched than a chain and less branched than a star having the same number of vertices:

$$B_{\text{chain}} < B_{\text{tree}} < B_{\text{star}} \quad (54)$$

Rule 1.3. In a tree having a constant number of vertices, the branching increases when the number of branches connected to a given vertex increases owing to the decrease in length of a long branch attached to the same vertex:

$$B_{n_j+1, N_j-1}^{\text{tree}} > B_{n_j, N_j}^{\text{tree}} \quad (55)$$

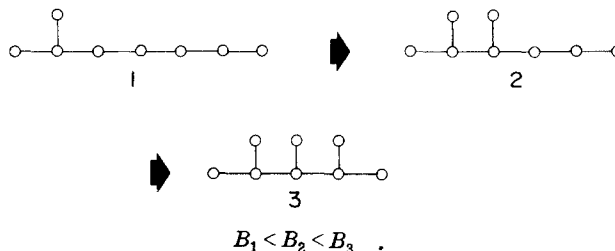
where the variable length of the branch is denoted by N_j and $N_j - 1$, respectively. It is also valid in the case when branches are attached likewise to other vertices of the tree:



The rules 1.1–1.3 show the combined influence of the total number of branches and the chain length (Rules 1.1 and 1.2) or the branch length (Rule 1.3) at a constant position of branching. Now keeping the length of branches constant but not their positions, we can formulate the following rule:

Rule 1.4. Among the trees having the same number of vertices and different number of branches of the same size attached to different vertices, the greater the number of branches the greater the branching:

$$B_{n_i, N_i=\text{const}}^{\text{tree}} < B_{n_i+1, N_i=\text{const}}^{\text{tree}} < B_{n_i+2, N_i=\text{const}}^{\text{tree}} < \dots \quad (56)$$



This inequality was proved earlier for branches with a minimal length only, but it could easily be extended to branches having arbitrary but equal length.

2. The influence of the chain length

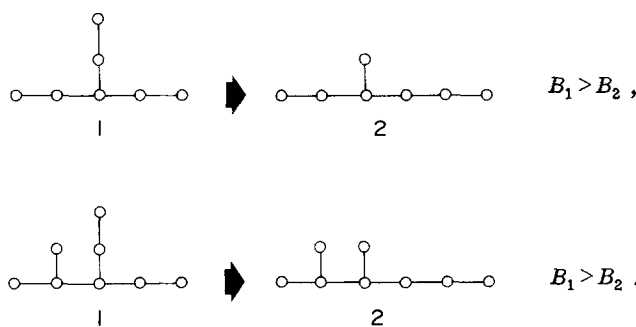
We have already formulated the influence of this factor on the branching when new branches are formed, attached to one vertex (Rules 1.1 and 1.2) or to different vertices (Rule 1.4). Another rule is formulated here for the case when the change in chain length is due to the change in the length of some branch.

Rule 2.1. The branching decreases when at a constant number of vertices the chain length increases at the cost of some branch length decrease.

This rule in fact reflects also the branch length influence:

$$B_{N_0+1, N_k-1}^{\text{tree}} < B_{N_0, N_k}^{\text{tree}} \quad (57)$$

Here N_k and $N_k - 1$ are the lengths of the branches which change; for example:



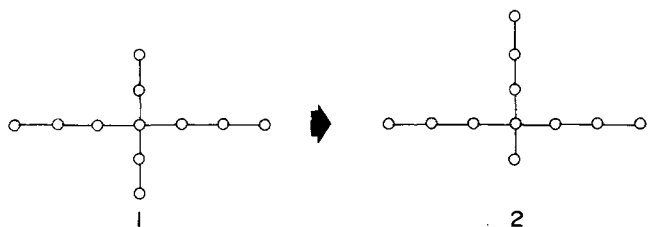
3. The influence of the branch length

It could be expressed in three separate rules, since one branch length may increase at the cost of a chain decrease or another branch decrease, or disappearing of another branch. The first and the third one were already formulated in Rule 2.1 and in Rule 1.3. They could be reformulated as follows: The branching increases when at a constant number of vertices in the graph some branch length increases at the cost of a chain decrease (Rule 2.1) or at the cost of a disappearance of some other branch (Rule 1.3). Only the following is new:

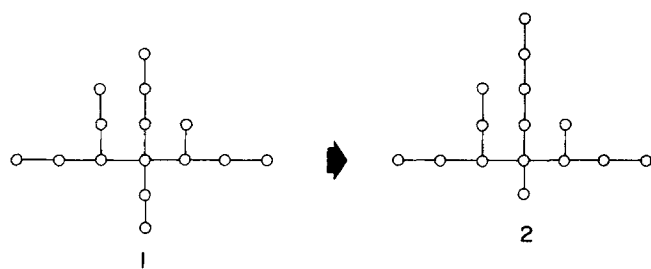
Rule 3.1. In graphs having a constant number of vertices, the branching decreases when some branch increases its length at the cost of the decrease of the length of another branch having the same or smaller size than the first branch:

$$B_{N_k+1, N_k-1}^{\text{tree}} < B_{N_k, N_k}^{\text{tree}}; \quad (58)$$

for example:



$$B_2 < B_1,$$



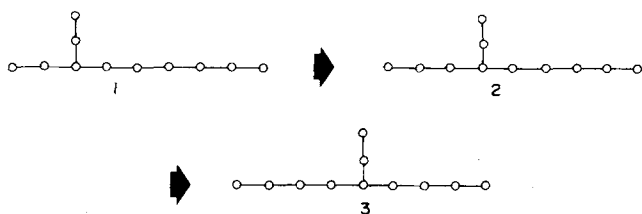
$$B_2 < B_1.$$

4. The influence of the branch position

Rule 4.1. In graphs having a constant number of vertices as well as a constant number and length of branches, the branching increases with the displacement of the branch from the terminal to the middle vertex of the chain:

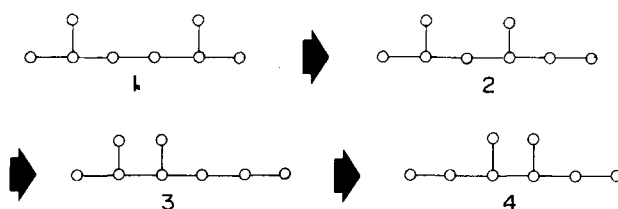
$$B_j^{\text{tree}} < B_{j+1}^{\text{tree}} < B_{j+2}^{\text{tree}} < \dots < B_{j=N_0/2}^{\text{tree}}. \quad (59)$$

This inequality was proved earlier for the trees having one branch only. The inequality gets stronger when the branch length increases; for example:



$$B_1 < B_2 < B_3.$$

In the case of more branched trees, it is difficult to formulate a simple rule, but a general tendency exists for trees to be more branched when more branches have positions near the middle of the chain [see Eqs. (28) and (29)]:



$$B_1 < B_2 < B_3 < B_4.$$

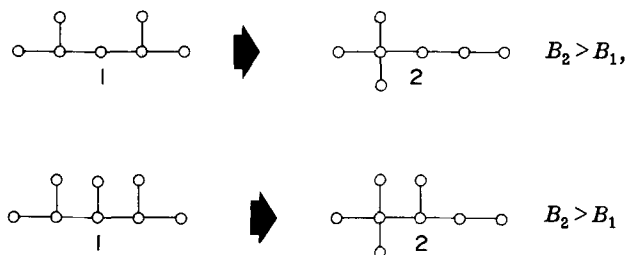
5. The influence of the number of branches attached to a given vertex, n_j

It was already shown in Rules 1.2 and 1.3 that the branching increases when the number of branches attached to a given vertex increases at the cost of the decrease of chain length or the length of a longer branch attached to the same vertex. In both cases, the total number of vertices, n , increases and these rules display the influence of n rather than that of n_j . The last factor should be examined at a constant total number of branches that means to examine the change in branching when the branches attached to different vertices are displaced for one vertex. Using the inequality (45), the following rule is formulated.

Rule 5.1. In a tree having a constant number of vertices and branches, the branching increases when a branch is displaced from one vertex to another branched one, having equivalent position in the chain:

$$B_{n_j+1, n_j-1}^{\text{tree}} > B_{n_j, n_j}^{\text{tree}}, \quad (60)$$

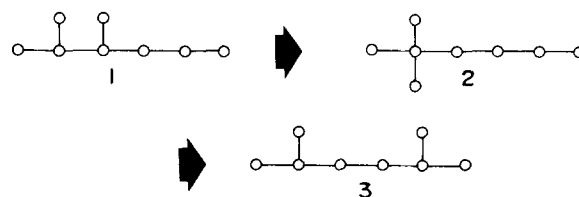
where $n_j = N_0 - j + 1$ is a vertex equivalent in a position to the vertex j . This rule was proved by inequality (46) for the simplest case of trees having only two branches of minimal length, but it could easily be generalized for an arbitrary branch length as well as for graphs having an arbitrary number of branches:



$$B_2 > B_1,$$

$$B_2 > B_1.$$

When replacing one branch if the initial and the last vertex are not equivalent in position, the branching could increase or decrease or even remain unchanged depending strongly on the vertex position [see inequality (45)]. That means the number of branches attached to a given vertex is a weaker factor than vertex position:

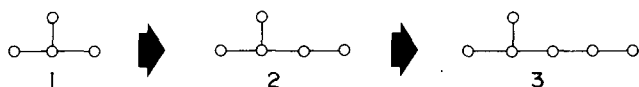


$$B_1 > B_2 > B_3.$$

B. $N \neq \text{const}$

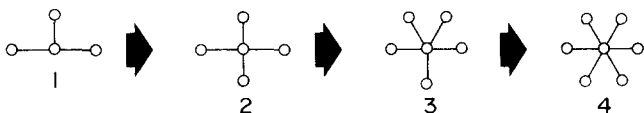
As it was mentioned in the analysis of Table II, there is no quantity among the six examined branching indices which could reflect all of the features of branching at the variable number of vertices of the graph. However, it is possible to reflect some of these on the basis of the average information on graph distances, $\bar{I}_D^E$.

Rule 6.1. *The relative branching decreases when the increase of the total number of vertices of the graph increases the chain length only. For instance,*



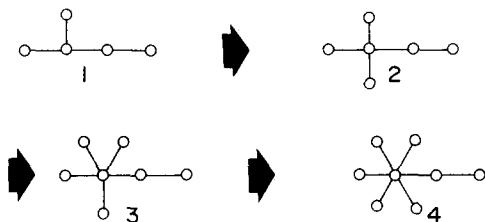
$$\bar{I}_D^E = 1 < 1.5219 < 1.9086 \text{ or } \bar{B}_1 > \bar{B}_2 > \bar{B}_3.$$

Rule 6.2. *The relative branching increases when the increase of the total number of vertices of the graph increases the number of branches attached to a given vertex. For instance:*



$$\bar{I}_D^E = 1 > 0.9710 > 0.9183 > 0.8631,$$

$$\text{or } \bar{B}_1 < \bar{B}_2 < \bar{B}_3 < \bar{B}_4;$$



$$\bar{I}_D^E = 1.5219 > 1.5059 > 1.4607 > 1.4052,$$

$$\text{or } \bar{B}_1 < \bar{B}_2 < \bar{B}_3 < \bar{B}_4.$$

VI. RULES ON BRANCHING AND MOLECULAR PROPERTIES

Every quantity used for describing the topological properties of graphs could be of importance for chemistry only if it satisfactorily reflects a number of chemical properties of the molecules. Related to these, numerous correlations have been reported between topological indices and molecular properties.^{11,15,35} Such correlations express well the dependence of a given molecular characteristic on the number of carbon atoms. However, the points most deviating from the correlation usually characterize the most branched structures. In this way, the branching of molecules is not satisfactorily reflected in correlations with different molecular properties when molecules having different numbers of carbon atoms are compared.

One way of avoiding these difficulties is in the use of two-parametric dependences. Wiener⁶ first expressed the boiling points of paraffins as a function of two parameters, the topological index W (Wiener number) and

the so-called polarity number. Other two parametric relations were reported by Hosoya,⁹ Randić,¹¹ and other authors.³⁵ This way usually leads to empirical formulas fitting the experimental data well, and it is of practical use for chemistry. It is no less important, however, to try to understand better the topological aspects of the molecular properties. Such an approach was developed by Hosoya,⁹ who defined 10 empirical rules on the boiling points of paraffin isomers.

Another approach is introduced in this paper. We first tried to understand the essence of branching on the appropriate theoretical basis and have expressed the main features of branching in a number of rules. In what follows, we shall demonstrate that our theoretical rules reflect in quite a satisfactory way a number of molecular properties of alkanes.³⁹ In order to check our rules, we show in Table III a necessary minimum amount of such experimental data.

Inspection of the data given in Table III and in Ref. 39 shows that, in principle, when the branching increases, the density d_4^{20} , the reflection coefficient n_D^{20} , and the heat of formation of gaseous alkanes from elements, $H_{\text{el gas}}^{25}$, increase, while the molecular volume V_M^{20} , the molecular refraction R_M^{20} , the latent heat of vaporization λ° , the heat of combustion in liquid state, $H_{\text{comb liq}}^{25}$, the boiling point T_B , and the vapor pressure coefficient b decrease. This tendency, however, is violated in some cases by the influence of an opposite acting factor. Wiener⁶ and Platt¹⁶ considered this second factor as an alternating polarity along the chain. Hosoya⁹ pointed out that the zigzag change in boiling point along the chain "reminds one of the melting points of chain compounds with the increase of the chain length." We shall prefer to speak in more general terms about the *metric factor* opposing, in some cases, the topological one. As a result of the action of this additional factor, our Rule 4.1 could not be confirmed by any of the examined nine molecular characteristics. Instead of a regular change of molecular properties when replacing the branch from the terminal to the middle vertex in the chain, as required by our Rule 4.1, an oscillating zigzag change occurs (see Table III). This causes deviations from some of our branching rules when the branch is on the next-to-the-terminal position (2 or $N_0 - 1$). These deviations, however, comprise only a part of the examined properties.

Rule 1.2 could be examined only in its first part, since no star graphs exist in carbon chemistry except for $N=4$ and 5, which are not sufficient. As a consequence of this rule, we can state that normal alkanes have maximal values of their boiling points (Hosoya⁹), latent heat of vaporization, the heat of combustion and the vapor pressure coefficient, and minimal value of their heat of formation of gaseous alkanes:

$$X_{\text{normal alkanes}} > X_{\text{branched alkanes}} \quad (X \equiv \Delta H_{\text{comb liq}}^{25}, T_B, \lambda^\circ, b),$$

$$X'_{\text{normal alkanes}} < X'_{\text{branched alkanes}} \quad (X' \equiv \Delta H_{\text{el gas}}^{25}).$$

The other four examined properties do not obey this rule. 2-Methyl substituents have anomalous large values (V_M^{20} , R_M^{20}) or anomalous small values (d_4^{20} , n_D^{20}), respectively, greater or smaller than those of the normal

TABLE III. Branching and molecular properties of alkanes having seven and eight carbon atoms. ^a

Structures ^b	T_B	λ°	V_M^{20}	R_M^{20}	b	$\Delta H_{el}^{25} \text{ gas}$	$\Delta H_{comb}^{25} \text{ liq}$	d_4^{20}	n_D^{20}
$N_0 = 7$									
12.	98.4	1839.3	146.56	34.55	7.8388	-44.89	-1151.27	0.68366	1.38756
13.	90.1	1759.6	147.66	34.59	7.7289	-46.60	-1149.97	0.67859	1.38485
15.	80.5	1670.8	148.95	34.62	7.6048	-48.30	-1148.73	0.67270	1.38145
17.	79.2	1648.3	148.69	34.62	7.5579	-49.29	-1147.85	0.67385	1.38215
14.	91.9	...	145.82	34.46	...	-45.96	-1150.55	0.68713	1.38864
16.	93.5	1776.8	143.52	34.28	7.7319	-45.34	-1151.13	0.69816	1.39339
19.	86.1	1677.9	144.53	34.32	7.5530	-48.17	-1148.83	0.69327	1.39082
$N_0 = 8$									
23.	125.7	2075.2	162.59	39.19	8.1186	-49.82	-1307.53	0.70252	1.39743
24.	117.6	1985.6	163.66	39.23	7.9852	-51.50	-1306.28	0.69792	1.39494
26.	109.1	1901.1	164.70	39.26	7.8700	-53.21	-1305.00	0.69354	1.39246
28.	106.8	1871.8	164.28	39.25	7.8207	-53.71	-1304.64	0.69528	1.39349
25.	118.9	1995.6	161.83	39.10	7.9959	-50.84	-1306.92	0.70582	1.39848
29.	118.5	1987.6	160.07	38.94	7.9802	-50.40	-1307.39	0.71358	1.40162
33.	112.0	1889.5	160.88	38.99	7.8038	-52.61	-1305.68	0.71000	1.40009
31.	115.6	1946.5	160.50	38.98	7.9090	-51.13	-1306.86	0.71214	1.40111
36.	113.5	1898.7	158.89	38.87	7.8097	-51.97	-1306.28	0.71906	1.40422

^a T_B in °C; λ° in cal/mole; V_M^{20} and R_M^{20} in ml/mole; ΔH in kcal/mole; d_4^{20} in g/ml. λ° and b are taken from vapor pressure temperature dependence: $\ln p = -\lambda^\circ/T + b$.

^bEach number indicates an alkane represented by the corresponding tree graph in Fig. 1.

alkanes. It should be noted again that these deviations from our Rule 1.2 are not associated with the branching at all, since every conception of branching considers chains as less branched than trees. As pointed out above, these deviations are caused by metric rather than by topological factors.

Rule 1.1 is reflected in many more molecular properties than Rule 1.2. The following inequalities are valid for alkanes:

$$X_{n\text{-alkanes}} > X_{2\text{-methylalkanes}} > X_{2,2\text{-dimethylalkanes}}$$

$$(X \equiv \Delta H_{comb}^{25} \text{ liq}, T_B, \lambda^\circ, b),$$

$$X'_{n\text{-alkanes}} < X'_{2\text{-methylalkanes}} < X'_{2,2\text{-dimethylalkanes}}$$

$$(X' \equiv \Delta H_{el}^{25} \text{ gas}).$$

The density and reflection coefficient could also be added to X quantities, and molecular volume and molecular refraction to X' quantities, since they obey these two inequalities. The change of these four quantities, however, is in a direction opposite to that required by the rules on branching. Again the branching factor has less influence on these four quantities. However, if we take 3- or 4-substituted alkanes, all nine quantities change in agreement with Rule 1.1:

$$X_{n\text{-alkanes}} > X_{3\text{-methylalkanes}} > X_{3,3\text{-dimethylalkanes}}$$

$$(X \equiv \Delta H_{comb}^{25} \text{ liq}, T_B, \lambda^\circ, b, V_M^{20}, R_M^{20}),$$

$$X'_{n\text{-alkanes}} < X'_{3\text{-methylalkanes}} < X'_{3,3\text{-dimethylalkanes}}$$

$$(X' \equiv \Delta H_{el}^{25} \text{ gas}, d_4^{20}, n_d^{20}).$$

Rule 1.3, like Rule 1.2, is in agreement with the changes of the same five from nine quantities (see Table III):

$$X_{3\text{-ethylalkanes}} > X_{3,3\text{-dimethylalkanes}} \quad (X \equiv \Delta H_{el}^{25} \text{ gas}),$$

$$X'_{3\text{-ethylalkanes}} < X'_{3,3\text{-dimethylalkanes}} \quad (X' \equiv \Delta H_{comb}^{25} \text{ liq}, T_B, \lambda^\circ, b).$$

The influence of geometry factor is the strongest at 2-methylalkanes. This again causes some deviations from the requirements of Rule 1.4. If we write the inequalities for 2-substituted alkanes,

$$X_{n\text{-alkanes}} > X_{2\text{-methylalkanes}} > X_{2,3\text{-dimethylalkanes}}$$

$$> X_{2,3,4\text{-trimethylalkanes}},$$

$$X'_{n\text{-alkanes}} < X'_{2\text{-methylalkanes}} < X'_{2,3\text{-dimethylalkanes}}$$

$$< X'_{2,3,4\text{-trimethylalkanes}},$$

they will be in force for $X = T_B, \lambda^\circ, b$, and $\Delta H_{comb}^{25} \text{ liq}$ and $X' = \Delta H_{el}^{25} \text{ gas}$, respectively. The other four quantities ($d_4^{20}, n_D^{20}, V_M^{20}$, and R_M^{20}), however, will have their extreme values for 2-methyl substituents, but not for n -alkanes. If we consider, however, substituents at inner positions ($j=3, 4$), the Rule 1.4 will be reflected well by all nine examined molecular characteristics:

$$X_{n\text{-alkanes}} > X_{3\text{-methylalkanes}} > X_{2,3\text{-dimethylalkanes}}$$

$$> X_{2,3,4\text{-trimethylalkanes}},$$

$$X'_{n\text{-alkanes}} < X'_{3\text{-methylalkanes}} < X'_{2,3\text{-dimethylalkanes}}$$

$$< X'_{2,3,4\text{-trimethylalkanes}},$$

where

$$X \equiv T_B, \lambda^\circ, b, \Delta H_{comb}^{25} \text{ liq}, V_M^{20}, R_M^{20},$$

and

$$X' \equiv \Delta H_{el}^{25} \text{ gas}, d_4^{20}, n_d^{20}.$$

Substituents having two or more carbon atoms cannot be connected to the second vertex of the chain. In this way, all possible irregularities generated from the geometry factor could be avoided. Hence, one could expect Rule 2.1 will be valid for all examined properties:

$$X_{n\text{-alkanes}} > X_{3\text{-methylalkanes}} > X_{3\text{-ethylalkanes}},$$

$$X'_{n\text{-alkanes}} < X'_{3\text{-methylalkanes}} < X'_{3\text{-ethylalkanes}},$$

X and X' being as in the previous case of 3-substituted alkanes. In fact, the boiling point obeys this inequality only at $N_0 > 7$, but both heats of formation do not obey this inequality even for $N_0 = 9$.

The rule 3.1, which compares the branching in trees having several branches of different length connected to one vertex, could not be checked with the available experimental data³⁹ covering alkanes with less than 12 carbon atoms.

Finally, all nine molecular characteristics reflect Rule 5.1 well:

$$X_{2,2\text{-dimethylalkanes}} < X_{2,(n-1)\text{-dimethylalkanes}}$$

$$X'_{2,2\text{-dimethylalkanes}} > X'_{2,(n-1)\text{-dimethylalkanes}}$$

Here again, X and X' are the same as in the inequalities associated with Rule 1.4. Only the molecular refraction index seems to tend to very small changes, or to constant values.

In summary, we can say that from the eight branching rules, introduced in this paper for tree graphs having the same number of vertices, only one is not reflected in the nine examined molecular characteristics and one could not be checked because of the lack of experimental data. The other six branching rules are reflected quite satisfactorily in the molecular properties. Rules 1.3 and 5.1 are in agreement with experimental data for all nine studied quantities independently from the substitution position of the branches. The same is true for Rule 2.1 for seven quantities, excluding only the heats of formation and combustion. Rules 1.1 and 1.4 cover the experimental data for all nine molecular characteristics only in the case of substitution position different from that next to the terminal atom. When branching occurs on the 2- or $(N_0 - 1)$ position, these two rules cover the data for five quantities: the heat of combustion, the heat of formation of gaseous alkanes, the boiling point, the latent heat of vaporization, and the vapor pressure coefficient b . These quantities obey Rule 1.2 as well. In such a way they represent molecular properties depending essentially on branching. The other four quantities—the density, reflection coefficient, molecular volume, and molecular refraction—do not always obey the branching rules. In the cases of 2-methyl or 2,2-dimethyl, or 2, $(N_0 - 1)$ -dimethyl substitution, they are more influenced by the second nontopological factor. Further study of the nature of this factor and its influence on the molecular properties seems necessary.

VII. CONCLUDING REMARKS

In the last section of this paper the ability of the diverse topological measures, based on the distance matrix of the molecular graph, to elucidate a large number of structure-dependent molecular properties was demonstrated. Developed for the tree graphs initially, this approach will be extended in another work to molecular cyclic structures.⁴⁰ Such studies might make the problem clearer, i.e., show to what extent the connectivity (topology) defines the molecular properties. One should consider here not only branching and cyclization as topological factors influencing the molecular properties

greatly, but other factors which could also be of importance. First of all, there could be the alteration in the values of a number of properties along the main molecular chain. One could suppose that this is a metric rather than a topological factor because of the zigzag geometry of the chains of paraffins. But the fact that the Hosoya index¹⁸ based on the characteristic polynomial of the molecular graph indicates this alteration well seems to indicate again the topological nature of this effect. Nevertheless, the question of a comparative role of topology and metrics in determining molecular properties remains open.

The importance of the topological measures based on the distance matrix of the molecular graph in defining branching and the molecular properties may be vindicated by the following. The greater branching corresponds to the shorter interatomic distances in the molecule, to more homogeneous distance length distribution, and to a great compactness of the molecules. This is significant for intramolecular processes such as chemical reactions, melting and vaporization, etc. It is also expected to influence greatly the intramolecular effects, such as inner charge transfer or polarization, delocalization of electrons, the chemical reactivity of the atoms in the molecules, etc.

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